

Harshit Batra

+91 8750440068 | contactharshitbatra@gmail.com | schwarzschildkugelblitz | harshit-batra | Harshit Batra

Education

Netaji Subhas University of Technology

New Delhi, India

Bachelor of Technology in Computer Science and Engineering (Specialization in Data Science)

2020 - 2024

Relevant Coursework:

- Software:** Computer Programming (C & Python 3), Web Technology (HTML, CSS, JavaScript & Ruby on Rails), Data Structures, Design & Analysis of Algorithms, Operating Systems, Big Data Analytics (Apache Spark, Hadoop), Probability and Stochastic Processes, Soft Computing, Data Mining, Data Handling & Visualization Tools, Multimedia Analytics, Distributed Computing, Social Media Analytics, Natural Language Processing, Distributed Databases, Business Intelligence, Information Retrieval
- Hardware:** Electrical & Electronics Engineering, Digital Logic Design, Computer Architecture, Microprocessor & Microcontroller, Data Communication
- Clubs and Activities:** • Research Head at Indian Game Theory Society • Volunteer at Covid Care Centre • Event Organizer Moksha 2021

Summary of Skills

Programming Languages	C/ C++, Python, R, SQL
Frameworks and Libraries	TensorFlow, PyTorch, Keras, Numpy, Pandas, Scikit-learn, XGBoost, LightGBM, NLTK, spaCy, Hugging Face Transformers
Tools and Technologies	Git, Docker, Kubernetes, Jupyter Notebook, Visual Studio, Linux/Unix Shell Shell, LaTeX, ChatGPT
Soft Skills	Time Management, Teamwork, Problem-solving, Documentation

Work Experience

Zanskar Research

Bangalore

Machine Learning Quant

Dec 2023 - Present

- Engineered **data pipelines using C++** to efficiently process over **100 million NSE trading records**, and created **Python wrappers** for these C++ algorithms, enhancing data processing computational **efficiency by 43.4%**.
- Developed a library for **mining alphas** from volume data using **Genetic Algorithms** on mathematical expressions, improving **efficiency by 27%**.
- Utilized **XGBoost** to predict the likelihood of order cancellations, achieving around **96% accuracy** in distinguishing between real and fake liquidity.
- Applied **PCA** to analyze market correlations among different sets of stocks over different time-span, enhancing the understanding of market dynamics

WorldQuant

Remote

Research Consultant

August 2023 - December 2023

- Developed **market-neutral alphas**, for equities in the USA and China. Utilized an integrated approach, analyzing price, volume, and fundamental data to **design and refine signals**, ensuring they're responsive and robust in various market conditions.
- Optimized in-sample alpha performance**, achieving significant improvements in key metrics such as the **sharpe ratio** and **drawdown**.

RTI International

Remote

Consultant

June 2023- December 2023

- Developed and implemented methodologies for calculating **climate indicators (extreme heat, flood, and drought)** at block and census levels across the United States using **LOCAv2**, accounting for **spatial and temporal variability**.
- Instituted rigorous quality tests to ensure the accuracy and consistency of climate indicator calculations, enhancing data reliability and decision-making.

Basys.ai

Remote

Data Science Intern

May 2023 - July 2023

- Performed **data wrangling**, including **data imputation, normalization, and aggregation** to clean and transform longitudinal patient data
- Developed predictive models using **Random Forest and Gradient Boosting algorithms**, identified key features using **Recursive Feature Elimination (RFE) & LASSO Regularization**, to uncover patterns and trends within the patient data. Achieved a **2.5% improvement** in predictive accuracy

Manchester Metropolitan University

Remote

Data Science Intern

March 2023 - May 2023

- Developed a **multimodal model** that combines **Bio_ClinicalBERT, Vision Transformer (ViT), and Mel spectrograms** using a **Fusion Network** to **detect personality traits** from video clips. Utilized the **ChaLearn dataset with 10,000 clips from 3,000+ YouTube HD videos**.
- Achieved **85% accuracy** in evaluating traits such as extroversion and agreeableness, demonstrating the effectiveness of model in personality detection.

Netaji Subhas University Of Technology

Dwarka, Delhi

Research Assistant

Mar 2022 - December 2023

- Developed and applied nature-inspired optimization algorithms, such as **Genetic Algorithms and Particle Swarm Optimization**, for **training neural networks, feature selection, and phishing detection**, achieving high accuracy and efficiency in these applications.
- Mentored engineering students** in research on metaheuristics, providing guidance on effective research proposals, data collection and analysis, and project completion, resulting in successful research outcomes and demonstrating strong **leadership and collaboration skills**.

Machine Intelligence Research Lab

Remote

Research Intern

Oct 2022 - Apr 2023

- Developed and fine-tuned wind **forecasting models** using decomposition techniques such as **EEMD and CEEMDAN**. Successfully applied the model to forecast wind speeds at various locations, resulting in improved accuracy and performance.
- Optimized machine learning models including **LSTM, ELM, and FNN** using evolutionary computing techniques such as Genetic Algorithm and Particle Swarm Optimization. Improved the accuracy and performance of the models to achieve better results.

Evident Capital

Remote

Analyst Intern

Jan 2023 - Mar 2023

- Conducted extensive research and analysis of carbon markets, including **blockchain-based carbon credit trading**, by interpreting data from government reports, industry publications, and financial data to identify trends and investment opportunities.
- Analyzed policy developments** and navigated complex regulations to ensure compliance in carbon markets, developing expertise in **India's carbon market policies** and blockchain-based carbon credit trading.
- Utilized analytical skills to identify risks and opportunities, providing insights and recommendations to the investment manager.

Northeastern University

Remote

Research Intern

April 2022 - March 2023

- Developed and applied a methodology for assessing mental health issues through **social media analysis**, utilizing **web scraping** techniques to collect Twitter data. Successfully identified and characterized mental health issues such as depression, anxiety, and suicidal ideation.
- Employed **Latent Dirichlet Allocation (LDA)** for **topic modelling** and **Louvain algorithm** for **community detection**, analyzing social media data to identify latent topics and groups of users. Successfully identified and characterized various topics and communities within the data.

Centre for Embedded Product Design

Dwarka, Delhi

Intern

Aug 2021 - Aug 2022

- Developed educational aids for blind and visually impaired students, creating **tactile diagrams and interactive models** to enhance learning. Demonstrated **strong problem-solving** and compassionate skills.
- Assisted in TI-CEPD Lab's operations, **designing and testing circuits**, troubleshooting technical issues, and **mentoring over 25 participants** in the Advanced Faculty Development Program, focusing on **PCB design and fabrication**.

Tech Swarm

Remote

Embedded Engineer Intern

Apr 2021 - July 2021

- Developed efficient and reliable deployment nest for **Ari 5G and Nano Drones** that included **automatic loading and unloading, battery charging, and built-in tracking system** to monitor drone location and status.
- Programmed drones for autonomous flight tasks such as mapping, surveying, and delivery, enabling navigation in GPS-denied environments and obstacle avoidance, showcasing **embedded system design** skills.

Projects

Risk Management System | NLP, Webscapring, Statistical modelling, Data visualization & analysis |

- Built a robust Financial Risk Management System leveraging **data analysis, Monte Carlo simulation, and VaR** to quantify and proactively manage investment risks. Real-time **API integration** and **sentiment analysis** of live news feeds ensured continuous risk assessment and informed decision-making.
- Technical Stack:** Python, Django, ReactJS, MongoDB, NLTK, textblob, Pandas, yfinance

The Hive |Computer vision, Robotics, PCB design, Embedded Systems |

- Implemented an autonomous robot sorting system with **computer vision** and **Path finding**, achieving a **3 packages per minute** sorting rate and real-time package tracking for optimized warehouse logistics.
- Technical Stack:** Python, Opencv, Arduino, C++, KiCAD

IPL Chatbot |NLP, LLMs, Prompt Engineering, Retrieval-Augmented Generation |

- Implemented IPL chatbot using **Large Language Model, RAG** and **chain-of-thought reasoning**. Engineered prompts to achieve **>95% accuracy** in natural language understanding with **Mistral 7B** model. Developed efficient backend with FastAPI and caching for **sub-second response times**.
- Technical Stack:** Python, Langchain, FastAPI, HTML, CSS, JavaScript

Evothresholding |Image Processing, Optimization algorithms |

- Developed EvoThreshold, an image thresholding framework for researchers integrating classical and modern **metaheuristic algorithms**. Implemented methods such as **Kapur's entropy** and **Otsu's thresholding**, for superior accuracy and efficiency in image segmentation tasks.
- Technical Stack:** Python

Achievements

2024	Top 100 Performers (13,000+ Teams) , IMC Prosperity Trading Challenge	Remote
2022	National Finalist , Smart India Hackathon (SIH) 2022	Ghaziabad, India
2022	National Finalist (Top 10) - Grid 3.0 Robotics Track (10,000+ Teams) , Flipkart GRID 3.0 Robotics Competition	Remote
2024	Top 30 Performers (2000+ participants) , AlphaGrep's AlphaVerse Premier League (Global Quant Championship)	Remote

Certification

2022	Machine Learning & Deep Learning in Python & R , Certificate-ID : UC-25bc16e2-5e5d-49bd-a0da-f5813c8985e7	Udemy
2022	Spatial Analysis & Geospatial Data Science in Python , Certificate-ID : UC-ef63a289-e420-4052-ba7c-19f6d4509750	Udemy
2022	Data Science Bootcamp 2022 , Certificate-ID : UC-d80dcc2e-6c39-42d9-8acf-662d62a0b287	Udemy
2022	Computer Vision with Python , Certificate-ID : UC-cf4985bd-843a-48e5-812a-0f5fddf6f685	Udemy
2022	Quantitative Finance with Python , Certificate-ID : UC-b12a2ea7-7b3b-415d-a56f-fcb1d3dfd2b7	Udemy
2023	CS120: Bitcoin for Developers I , Certificate-ID : 4174023725HB	Saylor Academy
2024	Time Series Analysis and Forecasting using Python , Certificate-ID : UC-ac092d23-a220-438d-bd4a-cd86fbb95826	Udemy

Publications

JOURNAL ARTICLES

An enhanced Golden ratio optimization method using different metal ratios for training of feed-forward neural networks (Under review)	
Harshit Batra, Vijay Kumar Bohat	
Expert systems with applications	
Phototropic Growth Algorithm: A Novel Metaheuristic Inspired from Phototropic Growth of Plants (Under review)	
Vijay Kumar Bohat, Fatma A. Hashim, Harshit Batra, Manan Suri, Mohamed Abd Elaziz	
Knowledge based systems	

CONFERENCE PROCEEDINGS

Dynamic Neighborhood-Based Grey Wolf Optimizer With Dimension Learning-Based Hunting And Lévy Flight	
Aditya Anand, Harshit Batra, Sukrit Kumar Syal	
2023 Intelligent Methods, Systems, and Applications (IMSA), 2023	

Talks/Workshop

21 -23 March 2023	Delivered workshop on HR analytics at NTPC Power Management Institute, utilizing Python as the primary analytics tool
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